6.4 Solution: (a) From Eq. (5.105), the magnetic induction inside the sphere is

$$\mathbf{B} = \frac{2}{3}\mu_0 \mathbf{M}.$$

Due to the rotation of the sphere, electric field is generated, which is given

$$\mathbf{E} = \mathbf{v} \times \mathbf{B},$$

with $\mathbf{v} = -\boldsymbol{\omega} \times \mathbf{r}$. (The negative sign can be understood from the argument leading to Eq. (5.142). In a coordinate frame that moves with a velocity $\mathbf{v}$ relative to a frame where there is a magnetic induction $\mathbf{B}$, then the electric field in the moving frame is $\mathbf{v} \times \mathbf{B}$. Here, the laboratory is moving relative to the rotating sphere with $-\boldsymbol{\omega} \times \mathbf{r}$.) Therefore,

$$\mathbf{E} = -\frac{2}{3}\mu_0(\boldsymbol{\omega} \times \mathbf{r}) \times \mathbf{M} = -\frac{2}{3}\mu_0 M\omega r\Big((\hat{z} \times \hat{r}) \times \hat{r}\Big) = -\frac{2}{3}\mu_0 M\omega \mathbf{r}_\perp,$$

where $\hat{r} = \hat{z} + \hat{r}_\perp$. The induced charge density can be found by Gauss law,

$$\rho = \epsilon_0 \nabla \cdot \mathbf{E} = -\frac{2}{3}\mu_0\epsilon_0 M\omega \nabla \cdot \mathbf{r}_\perp = -\frac{4}{3}\mu_0\epsilon_0 M\omega,$$

as $\nabla \cdot \mathbf{r}_\perp = 2$. Since

$$m = \frac{4}{3}\pi R^3 M,$$

we can also express the charge density as

$$\rho = -\frac{m\omega}{\pi c^2 R^3},$$

with $(\mu_0\epsilon_0)^{-1} = c^2$.

(b) The scalar potential outside the sphere satisfies the Laplace equation and can be expanded as

$$\Phi_{\text{out}} = \sum_{l=0}^{\infty} \frac{B_l}{r^{l+1}} P_l(\cos\theta).$$

Then, the electric field outside the sphere is

$$\mathbf{E}_{\text{out}} = -\nabla\Phi_{\text{out}} = -\hat{r}\frac{\partial\Phi_{\text{out}}}{\partial r} - \hat{\theta}\frac{1}{r}\frac{\partial\Phi_{\text{out}}}{\partial\theta} = \hat{r}\sum_{l=0}^{\infty}\frac{(l+1)B_l}{r^{l+2}}P_l(\cos\theta) - \hat{\theta}\sum_{l=0}^{\infty}\frac{B_l}{r^{l+2}}P_l^1(\cos\theta),$$

where we have used the fact that

$$\frac{dP_l(\cos\theta)}{d\theta} = P_l^1(\cos\theta).$$

On the other hand, the electric field inside the sphere is

$$\mathbf{E}_{\text{in}} = -\frac{2}{3}\mu_0 M\omega r\sin\theta\hat{r}_\perp = -\frac{2}{3}\mu_0 M\omega r\sin\theta(\hat{r}\sin\theta + \hat{\theta}\cos\theta).$$

By the continuity of tangential electric field, the $\hat{\theta}$-component, at the surface, we must have $B_l \equiv 0$ for $l \neq 2$, and

$$-\frac{2}{3}\mu_0 M\omega R\sin\theta\cos\theta = -\frac{B_2}{R^4}(-3\sin\theta\cos\theta),$$

or

$$B_2 = -\frac{2}{9}\mu_0 M\omega R^5.$$

Comparing with the expansion of scalar potential in spherical harmonics, Eq. (4.1),

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0}\sum_{l=0}^{\infty}\sum_{m=-l}^{l}\frac{4\pi}{2l+1}q_{lm}\frac{Y_{lm}(\theta,\phi)}{r^{l+1}},$$

we can see that the only nonzero multipole moment is q_{20}. Therefore,

$$\Phi(\mathbf{x}) = \frac{1}{5\epsilon_0}q_{20}\frac{Y_{20}(\theta,\phi)}{r^3} = \frac{1}{5\epsilon_0}\sqrt{\frac{5}{4\pi}}q_{20}\frac{P_2(\cos\theta)}{r^3} = B_2\frac{P_2(\cos\theta)}{r^3},$$

and

$$q_{20} = \sqrt{\frac{4\pi}{5}}5\epsilon_0 B_2.$$

From Eq. (4.6), we know

$$Q_{33} = 2\sqrt{\frac{4\pi}{5}}q_{20} = 8\pi\epsilon_0 B_2.$$

The tracelessness of the quadrupole moment requires that

$$Q_{11} = Q_{22} = -\frac{1}{2}Q_{33}.$$

Thus, we have found all the quadrupole moments, with

$$Q_{33} = 8\pi\epsilon_0 B_2 = -\frac{4m\omega R^2}{3c^2}.$$

(c) The charge density at the surface is

$$\sigma(\theta) = \epsilon_0\left(E_{\text{out},r} - E_{\text{in},r}\right)\Big|_{r=R} = \epsilon_0\left(\frac{3B_2}{R^4}P_2(\cos\theta) + \frac{2}{3}\mu_0 M\omega R\sin^2\theta\right).$$

Using the relation that

$$\sin^2\theta = \frac{2}{3}\Big(1 - P_2(\cos\theta)\Big),$$

we then have

$$\begin{aligned}
\sigma(\theta) &= \epsilon_0\Big(-\frac{2}{3}\mu_0 M\omega R P_2(\cos\theta) + \frac{2}{3}\mu_0 M\omega R\cdot\frac{2}{3}\big(1 - P_2(\cos\theta)\big)\Big)\\
&= \epsilon_0\Big(\frac{4}{9}\mu_0 M\omega R - \frac{10}{9}\mu_0 M\omega R P_2(\cos\theta)\Big)\\
&= \frac{4}{9}\mu_0\epsilon_0 M\omega R\left(1 - \frac{5}{2}P_2(\cos\theta)\right)\\
&= \frac{1}{4\pi R^2}\frac{4m\omega}{3c^2}\left(1 - \frac{5}{2}P_2(\cos\theta)\right).
\end{aligned}$$

(d) The potential difference from the equator to the pole can be determined by

$$\mathcal{E} = \left|\int_0^R \mathbf{E}_{\text{in}}\cdot d\mathbf{r}_\perp\right| = \frac{2}{3}\mu_0 M\omega\int_0^R r_\perp dr_\perp = \frac{1}{3}\mu_0 M\omega R^2 = \frac{\mu_0 m\omega}{4\pi R}.$$